

OUTCOMES-BASED EDUCATION (OBE) COURSE SYLLABUS IN
ITP1232 - INFORMATION ASSURANCE AND SECURITY 1
Date enhanced: August 2026

Academic Year/ Semester	AY 2025-2026, 1 st Semester		
University	Vision: A transformative green University that fosters social equity and sustainability through innovation, global engagement, and resilient island development. Mission: To deliver excellent instruction, advance research and innovation, and lead extension programs that promote inclusivity and meet the evolving needs of the island province and beyond. Goal: Uphold the tradition of excellence in instruction, research, extension and production geared towards a strong academic-industry interface for societal development. Core Values: Respect, Integrity, Social Responsibility, Excellence, Commitment Institutional Outcomes: • A <i>Professional</i> who is morally upright, socially responsible and globally employable • A <i>Leader and Innovator</i> who inspires others and is committed to social transformation and national development • An <i>Environmental Advocate</i> committed to research, extension and production initiatives		
Course Title	ITP1232 - Information Assurance and Security 1		
Course Credits	2 units (Lecture), 1 unit (Lab)		
Course Description	This course provides the foundations of information assurance and security from a business perspective. Topics covered include human factors, compliance with regulations, personnel security, risk assessment and ethical considerations.		
SDG Focus	SDG 9: Industry, Innovation, and Infrastructure SDG 4: Quality Education SDG 8: Decent Work and Economic Growth SDG 16: Peace, Justice, and Strong Institutions		
Contact Hours/Week	5 contact hours per week (90 hours in a semester)		
Days/Time	Wednesday 7:00am – 12:00pm		
Pre-requisite Course Code- Course Title	ITP 622 - Systems Integration and Architecture		
Program Outcomes	A. Articulate and discuss the latest developments in the specific field of practice.	Program Indicators	Research and present a case study on a recent enterprise security breach or an emerging threat (e.g., zero-day exploits, ransomware-as-a-service), evaluating the breakdown of the security models involved.
	B. Effectively communicate orally and in writing using both English and Filipino.		Draft clear, written Acceptable Use Policies (AUP) in English and deliver a simulated security awareness brief to human resource personnel using accessible localized terminology.
	D. Act in recognition of professional, social and ethical responsibility.		Evaluate an organizational scenario involving employee monitoring, separation of duties, or the handling of sensitive personnel data to ensure compliance with privacy responsibilities and professional ethical mandates.

	F. Apply knowledge of computing science, and mathematics appropriate to the discipline.		Utilize structural modeling concepts to design network defense architectures, establishing secure Zones of Trust and calculating resource availability constraints for backup solutions.
	G. Understand best practices and standards and their applications.		Map organizational physical security measures and disaster recovery frameworks against recognized international standard specifications and compliance guidelines.
	H. Analyze complex problems and identify and define the computing requirements appropriate to its solution.		Perform a formal physical vulnerability assessment of a target site infrastructure and define the technical computing specifications required for an automated physical intrusion detection system.
	I. Identify and analyze user needs and take them into account in the selection, creation, evaluation, and administration of computer-based systems.		Evaluate an organization's business processes versus technical controls to design an administrative security plan that accounts for user interaction, access needs, and staffing structures.
	J. Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.		Design and test a secure deployment plan for web or client-side applications that mitigates rising application threats within strict organizational development constraints.
	K. Integrate IT-based solutions into the user environment effectively.		Formulate a structured change management procedure that introduces vital security updates and patches into an active enterprise environment while minimizing operational friction.
	L. Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession.		Apply industry-standard risk analysis methods and defensive design models (such as the CIA Triad) to justify security architecture investments and protect critical infrastructure components.
	N. Assist in the creation of an effective IT project plan		Contribute to the generation of a comprehensive project roadmap for a multi-phased Information Security Program, scheduling strategic and tactical defense milestones.
	O. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.		Document and report a complex application vulnerability discovery, writing high-level executive summaries for corporate boards alongside detailed technical remediation protocols for the computing community.
	P. Analyze the local and global impact of computing information technology on individuals, organizations, and society.		Critically analyze how global supply chain disruptions or sudden systemic software application failures ripple outward to impact local business continuity and societal infrastructure.
	Q. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.		Outline the legal constraints and professional protocol mandates for when an organization must involve law enforcement agencies during a critical security breach incident.

	R. Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.		Research, list, and compare specialized security credentials (such as professional security certifications) to map out a clear personal path for staying current with evolving security tools and professional expectations.
	S. Participate in the generation of new knowledge or in research and development projects		Design, test, and document a robust contingency planning framework that integrates novel approaches to high availability, data backup, and swift incident response.
	T. Support local, regional, and national development plans along education, environment, socio economic, health, gender and development, science, and technology.		Develop an educational security outreach plan or public sector Disaster Recovery Plan (DRP) structured to protect the continuity of services in local communities or regional centers.
Course Outcomes	Discuss policies and practices to systems integration and architecture to ensure secure system operation and information assurance.	Course Indicators	Produces a comparative concept map or written analysis accurately linking major developmental theories to specific teaching and learning implications.
	Perform a vulnerability analysis of a system and explain how design implementation, and installation of hardware and software contribute to vulnerabilities of the organization		Conduct a comprehensive security assessment of a system to identify and document how design flaws, improper software configuration, and weak physical hardware installations introduce organizational risks.
	Propose strategies on how to counterattack threats		Design and configure secure application architectures, network defense perimeters, and segmented Zones of Trust based on the CIA Triad to actively block potential cyber exploits.

Course Outline and Timeframe

Week 1-2	Course orientation, syllabus review, and Unit A.1: Key terms; Unit A.2: Information and Security Attack and Vulnerabilities Unit A.3: Anatomy of an Attack, and Unit A.4: Awareness and Management Commitment to security.
Week 3	Unit A.5: Security Policy; Unit Wrap-up and assessment for Unit A & Unit B.1: Data Centric Security and, Unit B. 2: Data-Masking
Week 4	Unit B. 3: Cyber Security Standards, Unit B. 4: Global Surveillance and Computer Security
Week 5	Unit Wrap-up & Unit C. 1 IT Risks
Week 6	Unit C. 1: IT Risk (Continuation); Case Study Chapter 1 submission
Week 7	Unit C.2: Cybercrime
Week 8	Unit C. 3: ITIL Security and Management and Unit Wrap-up and assessment &
Week 9	Comprehensive Review of Units A to D; MIDTERM EXAMINATION (Written assessment covering Units A to C).
Week 10	Unit D. 1: Cryptography, and Unit D. 2: Enterprise Information Security Architecture
Week 11	Unit D. 3: Network Security; Unit D. 4: Digital Risk Management; and Unit D. 5: Copyright Infringement
Week 12	Unit Wrap-up and assessment for Unit D & Unit E. 1: Confidentiality; and Unit E. 2: Data Integrity
Week 13	Unit E. 3: Authentication; and Unit E. 4: Non-Repudiation
Week 14	Unit E. 5: Symmetric Key; and Unit E. 6: Public Key Cryptography
Week 15	Unit E. 7: Cryptanalysis; and Unit E. 8: Strong Cryptography; Submission of IT Certification for Any cybersecurity certification
Week 16	Unit E. 9: Data Security; and Unit E.10: Network Security Service Unit Wrap-up and assessment for Unit E; Submission of Case Study for the Whole class
Week 17	Comprehensive Review of Units E to H; FINAL EXAMINATION (Written assessment covering Units D and E).

Alignment of Course Outcomes with Summary Assessment Tasks

Course Objectives	Summative Assessment Task	Details
CO1	Midterm Examination & Asset Vulnerability Assessment Project	A combination of a written

		examination (Week 8) testing Units A to C and a group project requiring a formal physical and structural vulnerability assessment of a target infrastructure (Week 14-15). Students document design flaws, configuration errors, and weak physical perimeter protections.
CO2	Secure Architecture Design Defense & Policy Framework Portfolio	A portfolio submission consisting of: (1) A drafted corporate Acceptable Use Policy (AUP) framework (Week 6), and (2) A structural modeling design map implementing segmented Zones of Trust, network defense perimeters, and mitigation protocols against OWASP Top 10 vulnerabilities (Week 10).
CO3	Final Examination & Strategic Disaster Recovery Plan (DRP) Case Study	A comprehensive written final examination (Week 16) covering Units D and E alongside a major case study report (Week 14) analyzing real-world business recovery. Students formulate a detailed DRP/BCP framework integrating backup schedules, high availability architectures, and structured change management rules.

Learning Plan							
Desired Learning Outcomes	Content Area	SDGs	Resources/ Materials	Teaching and Learning Activities*	Assessment Tasks	References	Time Frame and Content Hours
At the end of the lesson, the students shall be able to: - Describe the key terms associated with information assurance and security - Gain familiarity with the kinds of information security attacks and the corresponding vulnerabilities - Describe the anatomy of an attack and devise some security policies for the organization.	Information Assurance and Security: An Introduction - Key Terms - Information Security Attacks and Vulnerabilities - Anatomy of an Attack - Awareness and Management Commitment to Security - Security Policy	(SDG 16), Peace, Justice, Strong Institution	Nathan C., & Steven F. (2022) Human Aspects of Information Security and Assurance: 16th IFIP WG 11.12 International Symposium, HAISA 2022, Mytilene, Lesbos, Greece, July 6–8, 2022, ... and Communication Technology Book 658	- Lecture and classroom discussion - Brainstorming - Interactive discussion - Case Study - Guided Hands-on Activities Flexible learning (online and/or offline modality) mapping	- Graded Recitation - Quizzes - Assignment - Seat Work	- Content - Module - Discussion Tool - Laptop - Speaker - Microphone - Handouts - Computer/Laptop/ Cellphones Internet Connectivity	10 Hours (Week 1–3)
Identify the approaches necessary towards	B. Approaches and Protection Towards Information Security - Data-centric Security - Data Masking	SDG 16), Peace, Justice, Strong Institution	Peter S. (2023) Getting Small Businesses in Tune with Cybersecurity:	- Lecture and classroom discussion - Brainstorming	- Graded Recitation - Quizzes - Assignment	- Content - Module - Discussion Tool - Laptop	15 Hours (Week 3–5)

information security - Explain how these approaches are implemented - Analyze the organization's information security scenario and evaluate how these approaches would be utilized for securing and protecting IT systems on a real-time basis.	- Cyber Security Standards - Global Surveillance - Computer Security		Securing your business from cyber threats	- Interactive discussion - Case Study - Guided Hands-on Activities - Flexible learning (online and/or offline modality) in times of class disruptions	Seat Work	- Speaker - Microphone - Handouts - Computer/Laptop/Cellphones Internet Connectivity	
Describe the challenges and threats in IT field and the manner of managing them to ensure information security in the organization.	C. Challenges and Threats to Information Technology - IT Risks - Cybercrime - ITIL Security Management	SDG 16), Peace, Justice, Strong Institution	Darril G., & Andy I. (2021) Managing Risk in Information Systems (Information Systems Security & Assurance) 3rd Edition	- Lecture and classroom discussion - Brainstorming - Interactive discussion - Case Study - Guided Hands-on Activities Flexible learning (online and/or offline modality) in times of class disruptions	- Graded Recitation - Unit Quizzes - Policy Drafting Assignment - Seat Work	- Content - Module - Discussion Tool - Laptop - Speaker - Microphone - Handouts - Computer/Laptop/Cellphones Internet Connectivity	17.5 Hours (Week 6-8)
MIDTERM EXAMINATION		2.5 Hours – (Week 9)					
At the end of the lesson, the students shall be able to: - Explain how cryptography could help ensure information assurance in the organization; - Assess how cryptography could help mitigate the risks and vulnerabilities to information security in the organization	D. Cryptography: A Comprehensive Study - Cryptography - Enterprise Information Security Architecture - Network Security - Digital Rights Management - Copyright Infringement	SDG 16), Peace, Justice, Strong Institution	David W. (2021) Real-World Cryptography	- Lecture and classroom discussion - Brainstorming - Interactive discussion - Case Study - Guided Hands-on Activities - Flexible learning (online and/or offline modality) in times of class disruptions	- Graded Recitation - Vulnerability Quizzes - Threat Modeling Assignment - Seat Work	- OWASP Guidelines Handouts - PowerPoint Presentation - LCD Projector/TV - Laptop/Cellphones - Internet Connectivity	20 Hours (Week 10-11)
- Describe the various aspects and forms of cryptography that could be utilized to guarantee the information assurance in the organization - Make use of appropriate cryptography technique in mitigating the information security challenges and threats present in an organization	E. Various Aspects and Forms of Cryptography - Confidentiality - Data Integrity - Authentication - Non-Repudiation - Symmetric Key Algorithm - Public Key Cryptography - Cryptanalysis - Strong Cryptography - Data Security - Network Security Service	SDG 16), Peace, Justice, Strong Institution	David W. (2021) Real-World Cryptography	- Lecture & Process Walkthroughs - Operational Brainstorming - Interactive Discussion - Group Dynamics: Designing a Change Request Workflow	Graded Recitation - Quizzes - Assignment: Incident Reporting Protocol - Seat Work	- Content - Module - Discussion Tool - Laptop - Speaker - Microphone - Handouts - Computer/Laptop/Cellphones Internet Connectivity	22.5 Hours (Week 12-16)
FINAL EXAMINATION		2.5 Hours (Week 17)					

* Flexible learning modalities shall be implemented in the event of suspended face-to-face classes due to weather-related emergencies or other institutional interventions.

Suggested Additional References:

- [1] Sijia W. (2022) Basic Cybersecurity: For Small Businesses Ready to Protect Themselves Against Rising Cyber Threats
 [2] Erich B. (2022) Introduction To Threat Intelligence: Properly Evaluate Your Threat Intelligence Program

Course Requirements:

- Written outputs in this course such as assignments, quizzes, seatwork, and problem set or exercises, must be submitted before the end of the semester or special permission must be requested from the Instructor/Professor before the due date.
- Take Midterm and Final examinations at the campus

Grading System: (Per Board Resolution No. 3, 4 or 5, s. 2016)

The final grade of each student will be computed as follows:

Distribution of Weight for the Overall/Final Grade

Mid-Term Grade	50%
Tentative	<u>50%</u>
Final Grade	100%

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Mid-Term Grade	50%
Tentative	<u>50%</u>
Final Grade	100%

Computation of Scores	
Written Output	50%
Performance	50%
Midterm/Final Examination	100%
Computation of Scores	
Written Output	50%
Performance	50%
Midterm/Final Examination	100%

Course Assessment Measures: <i>Lecture:</i> To strengthen assessment measures, end-of-term student survey on course effectiveness, instructional strategies and learning experience is utilized for learner feedback, brief instructor reflections to guide continuous improvement, and standardized rubrics to ensure fair and consistent grading aligned with course outcomes. <i>Laboratory/Practical Course:</i> To strengthen assessment measures, an end-of-term student feedback survey on course effectiveness, equipment utilization, and the hands-on learning experience is utilized for learner input, structured instructor reviews of practical sessions for continuous refinement, and standardized rubrics guaranteeing objective, outcome-aligned grading.
Continuous Quality Improvement Measures: To strengthen continuous quality improvement, feedback and peer evaluations are routinely analyzed alongside instructor reflections, while course outcomes are regularly aligned with program goals, CHED standards, and industry trends to ensure ongoing academic relevance.
Consultation Time/Venue: KJMB – 1:00pm – 3:30pm - Monday

Prepared by:	Reviewed/Checked:	Approved:
 KENNETH JOHN M. BAUTISTA <i>Faculty</i> <i>Date:</i> _____	 ASTER VIVIEN C. VARGAS, MSIT <i>Department Chair</i> <i>Date:</i> _____	 MARIA CONCEPCION S. VERA, DIT <i>Dean</i> <i>Date:</i> _____

Note for the faculty:

- All course program outcomes must explicitly integrate at least two of the three domains of learning (Cognitive, Affective, Psychomotor).
- For laboratory/practical courses, the Psychomotor domain must be integrated into all listed program outcomes to reflect hands-on competency.
- Course requirements shall comprise attendance, long quizzes/short quizzes, midterm and final examinations, class participation, worksheets/reports, projects, and any additional coursework deemed necessary according to the nature of the course.
- Include all POs of the program.